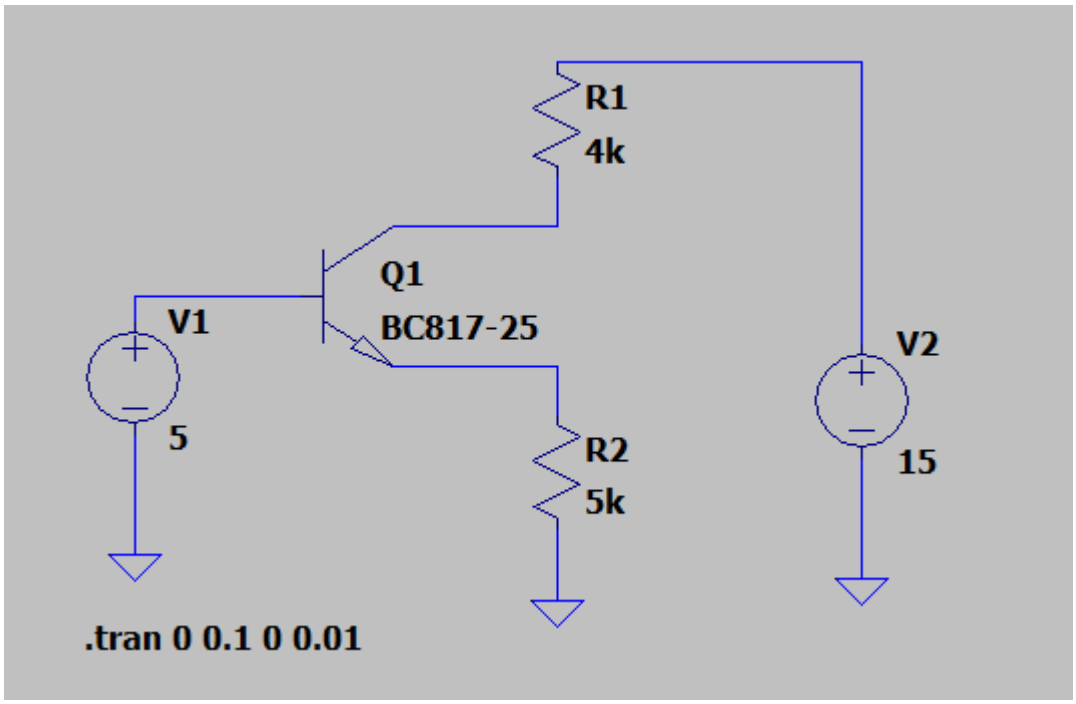
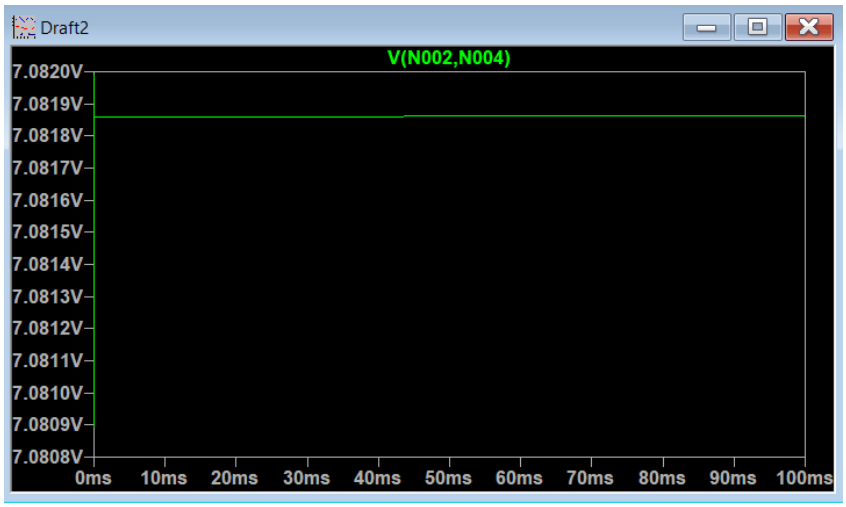
# Laborator 3 EEA

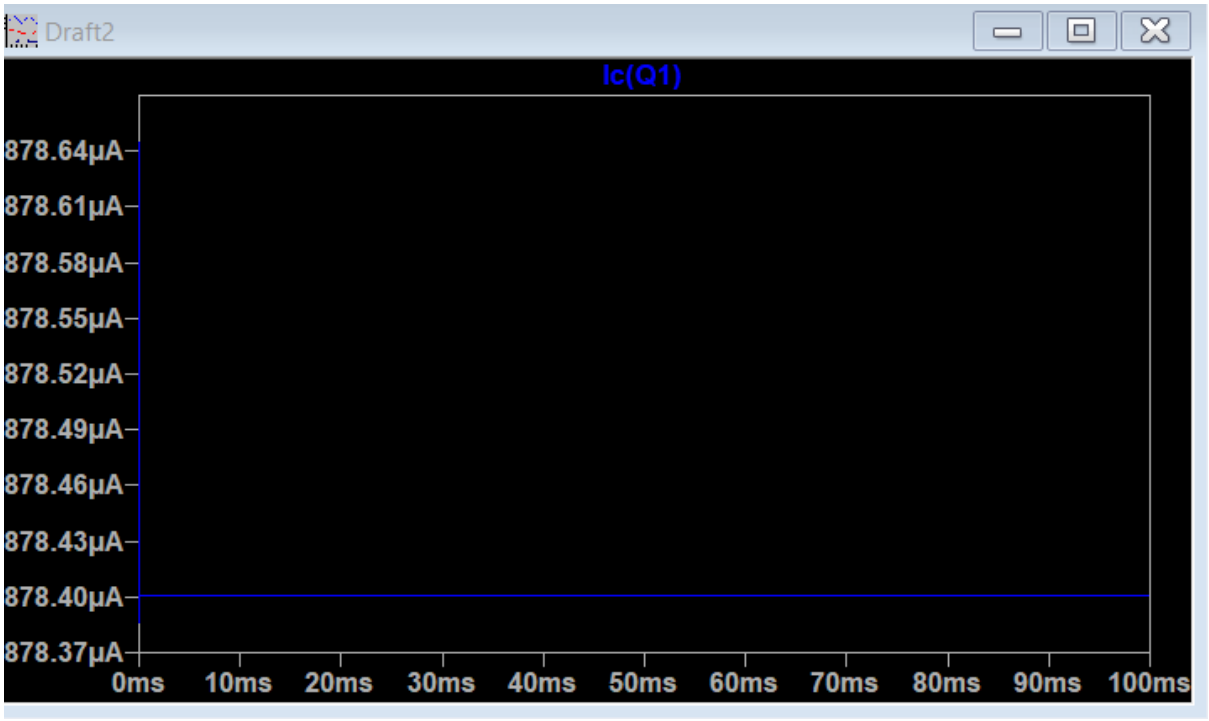
1. Tranzistor 1 = BC817-25



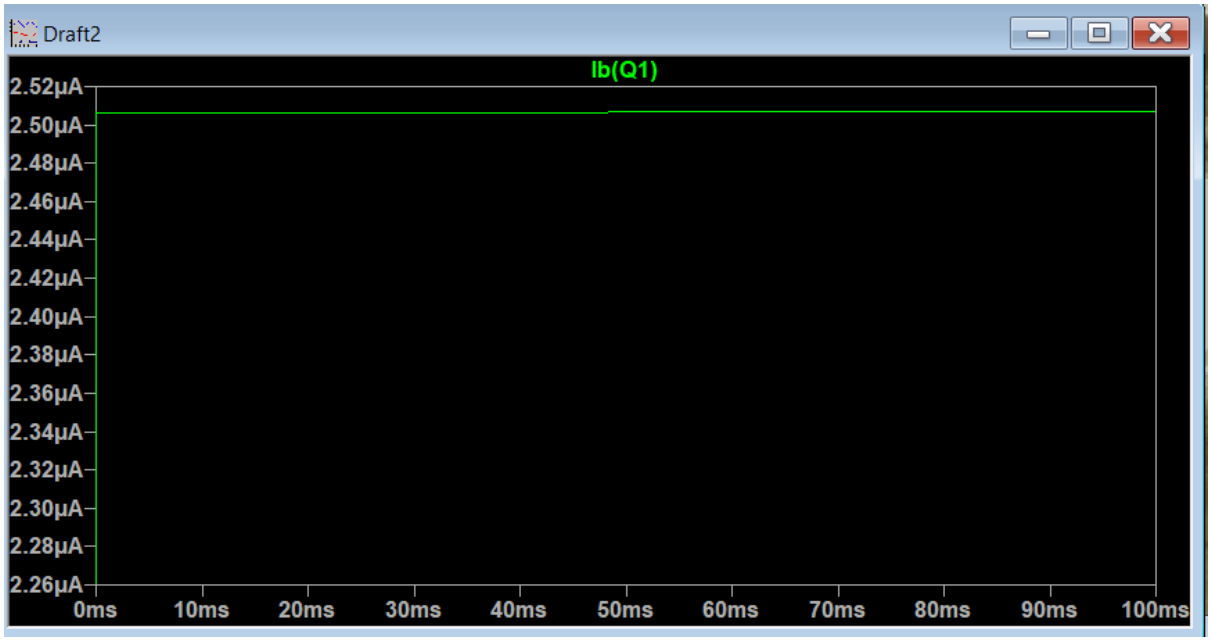

Se verifica corectitudinea UCE, conform graficului de mai jos.




Avem IC = 878.46 uA




Avem IB = 2.50 uA



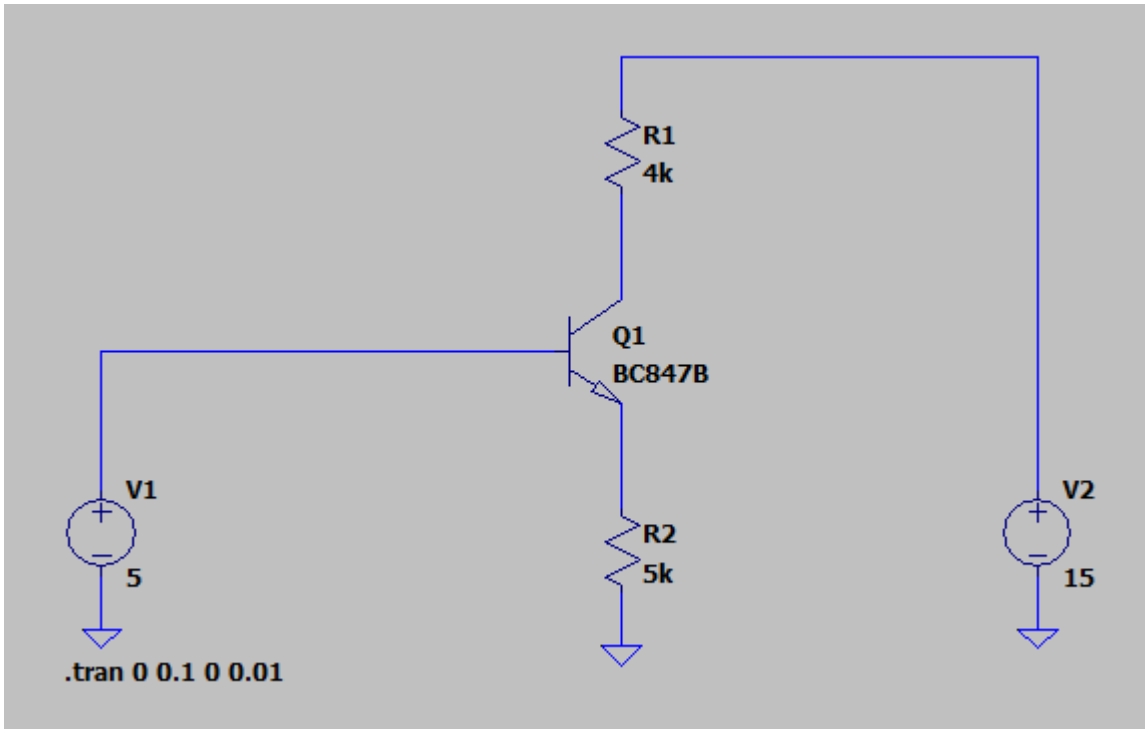

Beta este egal cu 350, deoarece beta = IC/IB. Valoarea este corectă, intrucat 350>100 conform datasheet-ului.

## Electrical Characteristics (@$T_A$ = +25°C, unless otherwise specified.)

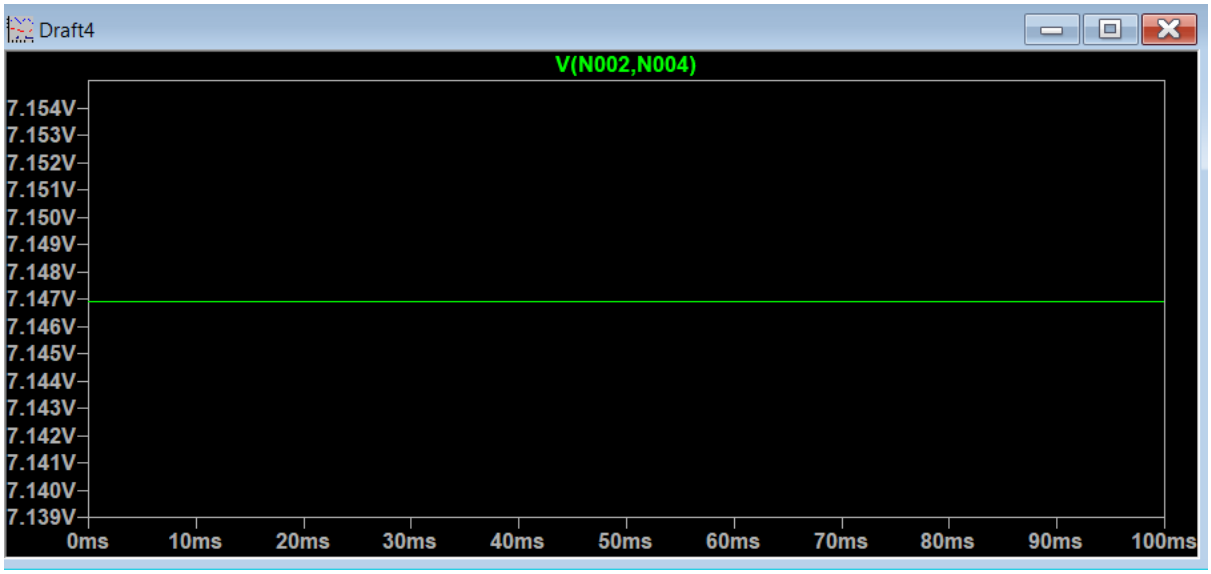
| Characteristic | | Symbol | Min | Typ | Max | Unit | Test Condition |
|---|---|---|---|---|---|---|---|
| Collector-Base Breakdown Voltage | | $BV_{CBO}$ | 50 | — | — | V | $I_C$ = 100μA |
| Collector-Emitter Breakdown Voltage | | $BV_{CEO}$ | 45 | — | — | V | $I_C$ = 10mA |
| Emitter-Base Breakdown Voltage | | $BV_{EBO}$ | 5 | — | — | V | $I_C$ = 100μA |
| Collector-Emitter Cut-Off Current | | $I_{CES}$ | — | — | 100<br>5.0 | nA<br>μA | $V_{CE}$ = 45V<br>$V_{CE}$ = 25V, $T_J$ = +150°C |
| Emitter-Base Cut-Off Current | | $I_{EBO}$ | — | — | 100 | nA | $V_{EB}$ = 5.0V |
| DC Current Gain (Note 9) | BC817-16<br>BC817-25<br>BC817-40 | $h_{FE}$ | 100<br>160<br>250 | — | 250<br>400<br>600 | — | $V_{CE}$ = 1.0V, $I_C$ = 100mA |
| | BC817-16<br>BC817-25<br>BC817-40 | | 60<br>100<br>170 | | — | | $V_{CE}$ = 1.0V, $I_C$ = 300mA |
| Collector-Emitter Saturation Voltage (Note 9) | | $V_{CE(SAT)}$ | — | — | 0.7 | V | $I_C$ = 500mA, $I_B$ = 50mA |
| Base-Emitter Voltage (Note 9) | | $V_{BE}$ | — | — | 1.2 | V | $V_{CE}$ = 1.0V, $I_C$ = 300mA |
| Gain Bandwidth Product | | $f_T$ | 100 | — | — | MHz | $V_{CE}$ = 5.0V, $I_C$ = 10mA, f = 50MHz |
| Collector-Base Capacitance | | $C_{CBO}$ | — | — | 12 | pF | $V_{CB}$ = 10V, f = 1.0MHz |

Note: 9. Measured under pulsed conditions. Pulse width ≤ 300μs. Duty cycle ≤ 2%.

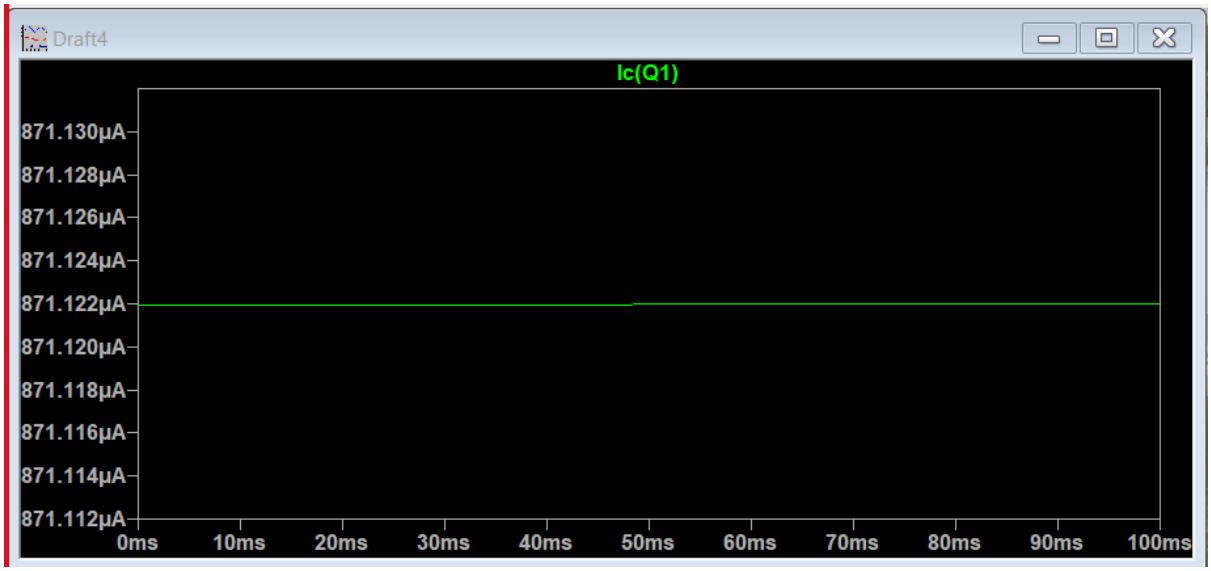
Tranzistor 2 = BC847B



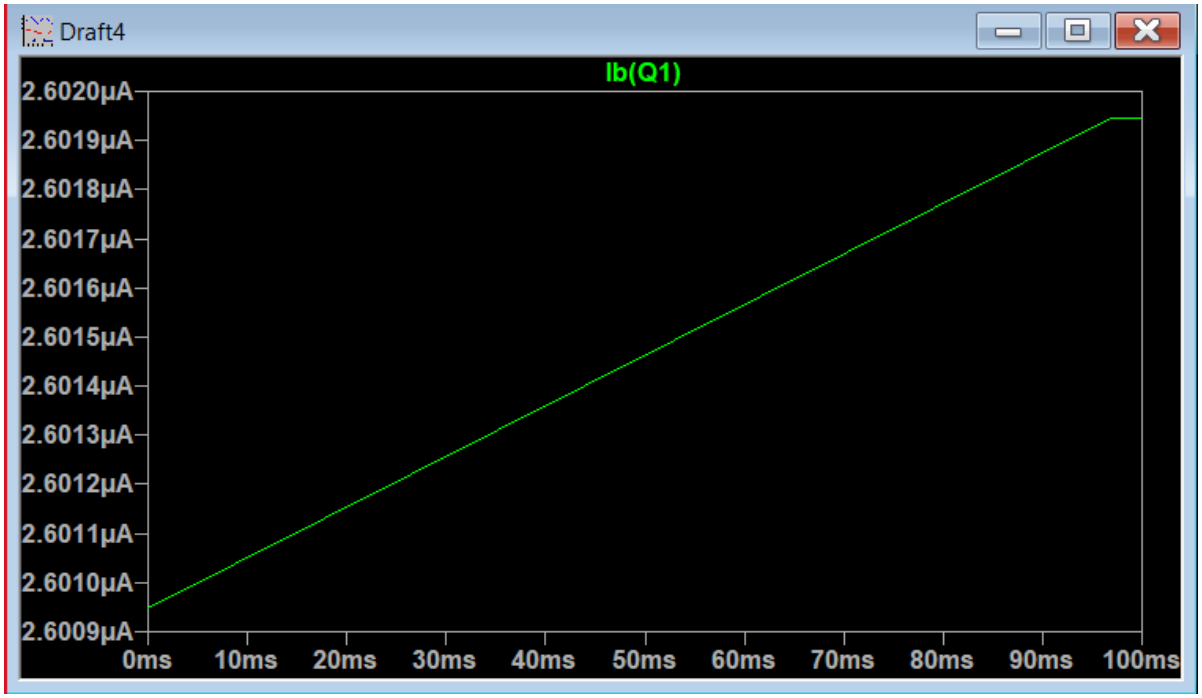

Se verifica corectitudinea UCE, conform graficului de mai jos.




Avem IC = 871.122uA




Avem IB = 2.601uA




Beta este egal cu 334,91 , deoarece beta = IC/IB. Valoarea este corectă, intrucat 200 < 334,91 < 450 conform datasheet-ului.

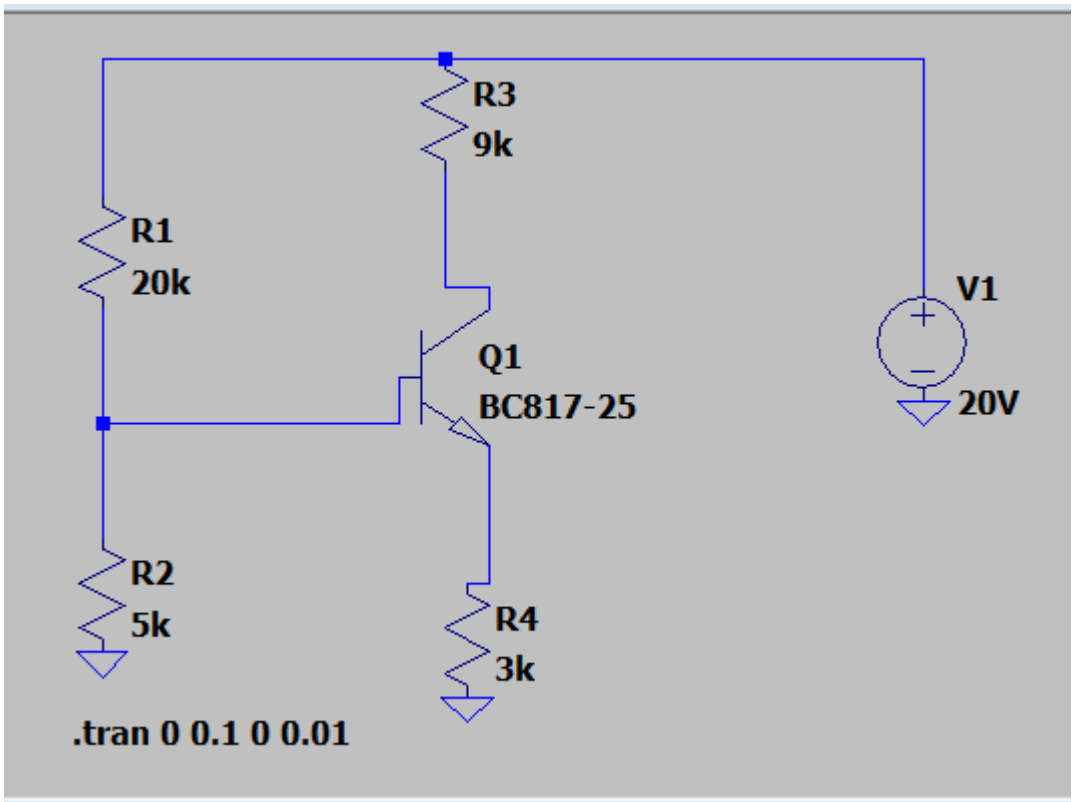
## 1.4. Quick reference data

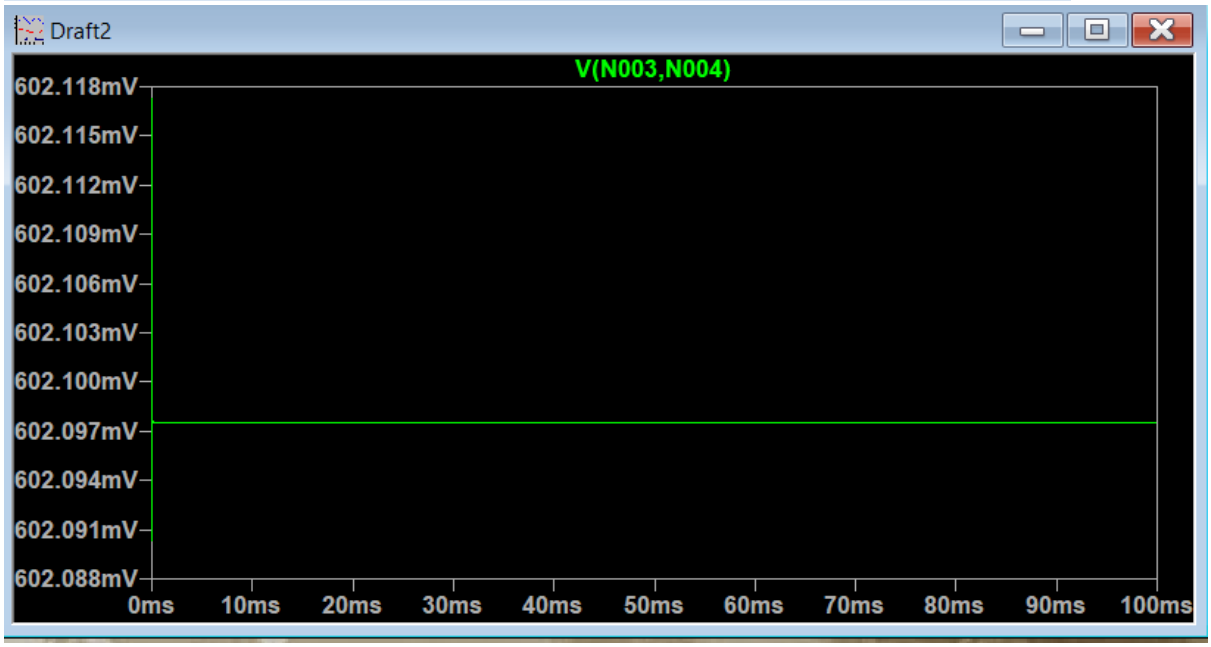
**Table 2. Quick reference data**

*$T_{amb}$ = 25 °C unless otherwise specified.*

| Symbol | Parameter | Conditions | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|
| $V_{CEO}$ | collector-emitter voltage | open base | - | - | 45 | V |
| $I_C$ | collector current | | - | - | 100 | mA |
| $h_{FE}$ | DC current gain | $V_{CE}$ = 5 V; $I_C$ = 2 mA | 110 | - | 800 | |
| | $h_{FE}$ group A | | 110 | 180 | 220 | |
| | $h_{FE}$ group B | | 200 | 290 | 450 | |
| | $h_{FE}$ group C | | 420 | 520 | 800 | |

2.


R3
9k
R1
20k
V1
Q1
BC817-25
20V
R2
5k
R4
3k
.tran 0 0.1 0 0.01


Draft2
V(N003,N004)
602.118mV
602.115mV
602.112mV
602.109mV
602.106mV
602.103mV
602.100mV
602.097mV
602.094mV
602.091mV
602.088mV
0ms
10ms
20ms
30ms
40ms
50ms
60ms
70ms
80ms
90ms
100ms

$I_C = 3\ mA$

$V_{CC} = 20\ V$

$U_{CE} = 8\ V$

$R_e = 1\ K\Omega$

$V_{CC} = U_{Rc} + U_{CE} + U_{Re}$

$U_{Rc} = V_{CC} - U_{CE} - U_{RE} = 20 - 8 - 3 = 9\ V$

$\Rightarrow R_C = 9\ K\Omega$